Breast Cancer Classification using Neural Networks

Project Synopsis: Naviotech Solution AI Internship (September 2026 batch)

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Abstract

Breast cancer is the most common cancer in women worldwide, and finding it early improves survival. This project builds a neural network in TensorFlow/Keras that classifies a breast tumour as **benign** or **malignant** from 30 measurements of cell nuclei taken from a fine-needle aspirate (FNA) image. Using the Breast Cancer Wisconsin (Diagnostic) dataset of 569 patients, the model reaches **96.49% accuracy** on unseen test patients, detects **41 of 42 cancers** (sensitivity 97.62%) and achieves a ROC-AUC of **0.994**. Five-fold cross-validation confirms a mean accuracy of **97.72% ± 0.89%**, the most consistent of five models compared.

Problem Statement

Given 30 numeric measurements describing the cell nuclei in an FNA sample of a breast mass, automatically classify the tumour as benign or malignant with high accuracy, and most importantly without missing cancers (false negatives).

Objectives

1. Develop an intelligent cancer prediction system based on a neural network.
2. Understand classification algorithms, including classical machine-learning baselines.
3. Learn healthcare data analysis: exploration, cleaning and preprocessing.
4. Apply neural networks to a real-world medical dataset.
5. Evaluate with accuracy, precision, recall, F1-score, confusion matrix and ROC-AUC.

Dataset

Property	Value
Name	Breast Cancer Wisconsin (Diagnostic), UCI ML Repository (via scikit-learn)
Patients	569: 212 malignant (37.3%) and 357 benign (62.7%)
Features	30 = 10 nucleus properties (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension) × mean, standard error, worst
Quality	No missing values, no duplicates

Methodology

Phase	Work done
1. Data collection	Load the dataset with scikit-learn
2. Data cleaning	Check missing values and duplicates
3. Preprocessing	Stratified split (364 train / 91 validation / 114 test); StandardScaler fitted on training data only
4. Network design	30 inputs → Dense 16 (ReLU) → Dropout 0.2 → Dense 8 (ReLU) → Dense 2 (Softmax); 650 parameters
5. Training	Adam (lr 0.001), sparse categorical cross-entropy, batch 32, early stopping (patience 10)
6. Evaluation	Accuracy, precision, recall, F1, specificity, confusion matrix, ROC-AUC, 5-fold CV, 4 baseline models
7. Prediction system	Notebook function and command-line tool for new patients

Tools and Technologies

Python 3 · TensorFlow / Keras · scikit-learn · NumPy · Pandas · Matplotlib · Seaborn · Google Colab / Jupyter Notebook · Git and GitHub

Results

Metric (114 test patients)	Value
Accuracy	96.49% (target > 90%)
Sensitivity / recall (malignant)	97.62%
Precision (malignant)	93.18%
F1-score (malignant)	95.35%
Specificity	95.83%
ROC-AUC	0.994
5-fold cross-validation accuracy	97.72% ± 0.89%

Applications

Medical diagnosis support systems · clinical decision-making and triage · healthcare analytics · disease prediction systems · research and development.

Conclusion

A small neural network classified breast tumours with high accuracy and very high sensitivity, well above the project target. One cancer was still missed with high confidence, so a model like this should only support a doctor's decision, not replace it. Classical models such as logistic regression and SVM did about as well on this small dataset, while the neural network gave the most consistent results across cross-validation folds. Future work could include larger multi-centre datasets, SHAP explanations, threshold tuning, image-based CNNs and a web application.